WAY4TECH SOLUTIONS

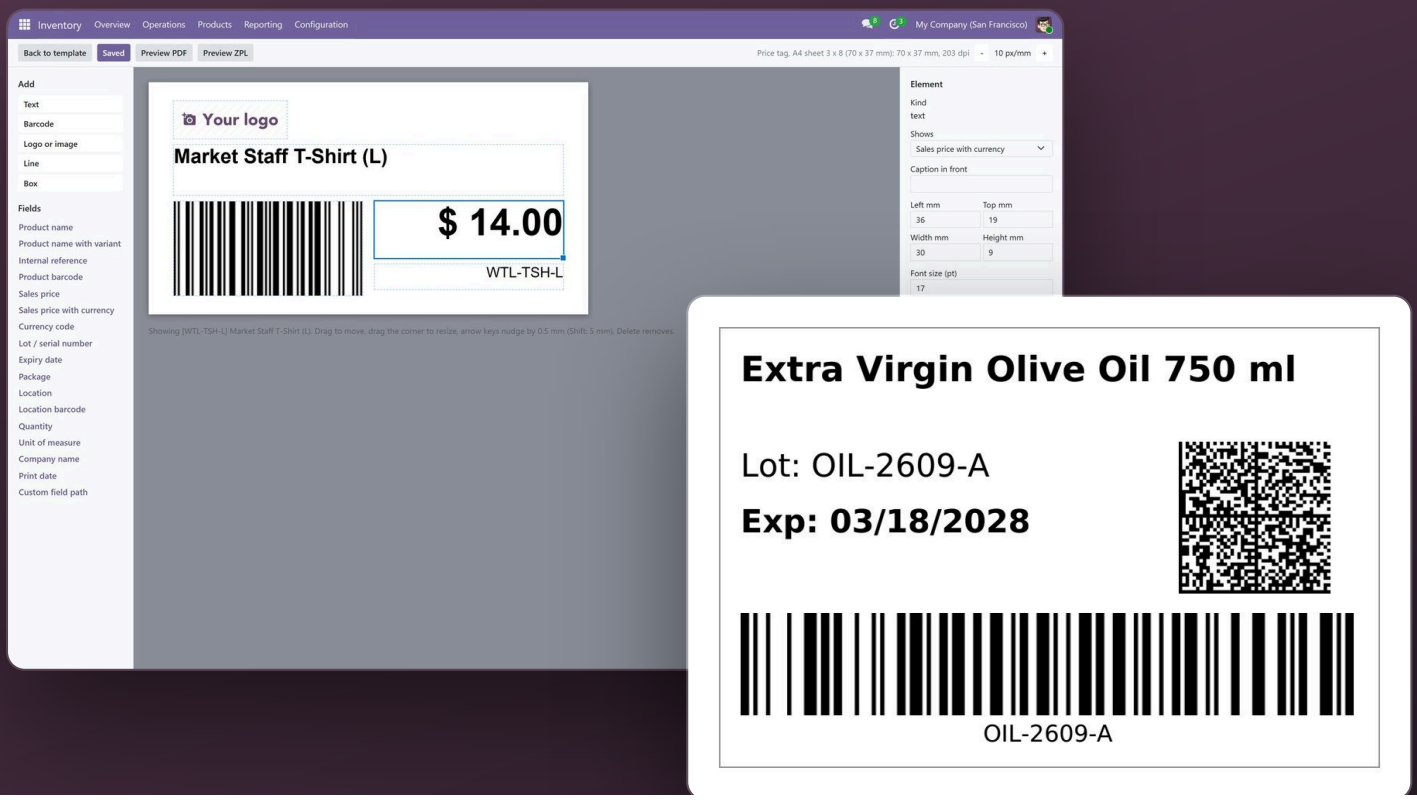
Modules
wt_label_designer · w4_direct_print
wt_print_suite · Odoo 19

COMPLETE GUIDE · DESIGN, PRINT AND SEND LABELS AND DOCUMENTS

Label Printing and Direct Print for Odoo 19

Design product, lot, package, location and price labels by drag and drop, print them as PDF sheets or Zebra ZPL, and send labels, invoices and delivery slips straight to the printer with no preview. Every screen and every step, no Odoo background needed.

LABEL DESIGNER | DIRECT PRINT | PRINT SUITE



START

About this guide

This guide takes you from an empty Odoo 19 database to labels that come out of the right printer by themselves. It covers three modules that work alone or together, one screen at a time. You do not need to have used Odoo before.

Who it is for

The warehouse or shop manager who sets labels up, the clerk who prints them at the receiving desk, the office that prints invoices, and whoever looks after the printers.

How to read it

Front to back the first time. Parts 3 to 7 are the Label Designer, parts 8 to 13 and 15 are Direct Print, part 14 is what the two do together. Each part opens with what you will learn and a short list of things to do.

The screenshots

Every screenshot was taken on Way4Tech's own test database for these modules, a small grocery and hardware shop with lot tracked goods, packages and shelf locations. The same shop is created by the modules' demo data on any test database.

The three badges

Each part and each feature carries a badge that names the module it needs.

BADGE	MODULE	WHAT IT GIVES YOU
LABEL DESIGNER	wt_label_designer	Label templates, the designer, the print wizard, PDF and ZPL output. Needs Inventory.
DIRECT PRINT	w4_direct_print, with the free wt_direct_print_stock and wt_direct_print_account	Printers, print agents, workstations, printing rules and the print queue. Any Odoo report prints with no preview.
PRINT SUITE	wt_print_suite (free)	Installs itself when both paid modules are present. Labels go straight to a printer, and labels print by themselves when a receipt is validated.

What is in the guide

PART 1 What the Print Suite does

PART 2 The menus at a glance

PART 3 Install and find your way in LABEL DESIGNER

PART 4 Set up a label template LABEL DESIGNER

PART 5 Design the label in the designer LABEL DESIGNER

PART 6 Print labels LABEL DESIGNER

PART 7 One design, two outputs: PDF and ZPL LABEL DESIGNER

PART 8 How Direct Print works DIRECT PRINT

PART 9 Settings and the print agent DIRECT PRINT

PART 10 Print agents and pairing DIRECT PRINT

PART 11 Printers DIRECT PRINT

PART 12 Workstations and personal preferences DIRECT PRINT

PART 13 Printing rules DIRECT PRINT

PART 14 Labels straight to the printer PRINT SUITE

PART 15 The print queue DIRECT PRINT

PART 16 Safety and what happens behind the scenes

PART 17 A fifteen minute walk-through

PART 18 Questions and fixes

PART 19 The menu map

PART 1 What the Print Suite does

Standard Odoo prints a few fixed product label layouts, always through a PDF preview that someone then sends to a printer by hand. That is fine for a sheet of price tags once a week. It is slow for a receiving desk that labels every carton, and it cannot drive a Zebra in its own language. These modules close both gaps.

IN THIS PART YOU WILL LEARN

- What each of the three modules does on its own
- What the two free companion modules add
- How a label travels from a design to a printer
- Which module you need for which job

The modules

MODULE	TECHNICAL NAME	PRICE	NEEDS
Label Designer	wt_label_designer	USD 59	Inventory (stock)
Direct Print	w4_direct_print	USD 79	Nothing beyond Odoo itself
Direct Print for Inventory	wt_direct_print_stock	Free	Installs itself with Direct Print and Inventory
Direct Print for Invoicing	wt_direct_print_account	Free	Installs itself with Direct Print and Invoicing
Print Suite	wt_print_suite	Free	Installs itself with Label Designer, Direct Print and Direct Print for Inventory

> The Odoo Apps Store adds the price of every paid module a module depends on, so the Print Suite listing shows the two paid modules together. There is no extra charge for the suite itself.

Label Designer on its own LABEL DESIGNER

You draw a label once: its size, whether it comes on a roll or an A4 or Letter sheet, and where the product name, price, barcode, lot number or logo sit. Then you print it from a product list, a receipt, a lot, a package or a location. The same design comes out as a PDF for any printer, or as ZPL, the language Zebra and compatible label printers speak, at 203, 300 or 600 dpi.

Direct Print on its own DIRECT PRINT

Every printer you own is set up once in Odoo. A printing rule says which report goes to which printer, for which person or desk. From then on, when someone clicks Print on an invoice or a delivery slip, it does not open a PDF preview; it goes straight to that printer. A small print agent on a computer in the building fetches the jobs over HTTPS, so Odoo can run in the cloud and nothing on your firewall has to change. With the free companions, documents can also print by themselves when a transfer is validated or an invoice is confirmed.

The two together PRINT SUITE

The print wizard gets a Send to printer button and a printer field, each template can name its default printer, and a printing rule can print labels for every line of a receipt the moment it is validated, one per unit or one per line.

From a design to a label in your hand

1. **Template** — Size, layout, dpi
2. **Designer** — Fields, barcodes, logo
3. **Print wizard** — Records and counts
4. **PDF or ZPL** — Download, or queue a job
5. **Agent or server** — Sends it to the printer

Which module for which job

YOU WANT TO	YOU NEED
Print shelf labels and price tags on A4 sheets from your laser printer	LABEL DESIGNER
Print lot labels with a Data Matrix and the expiry date on a Zebra roll	LABEL DESIGNER, ZPL downloaded and sent by hand, or PRINT SUITE to send it straight away
Click Print on an invoice and have it come out at the front office with no preview	DIRECT PRINT
Print the delivery slip when a delivery is validated	DIRECT PRINT (Inventory companion)
Print the invoice when it is confirmed	DIRECT PRINT (Invoicing companion)
Print one label per unit received the moment a receipt is validated	PRINT SUITE

PART 2 The menus at a glance

The Label Designer lives inside Inventory. Direct Print adds its own Printing app to the main menu. Here is where each screen sits; the same map is repeated at the end of the guide.

Inventory, Configuration, Labels

Label Templates	Every label design: size, layout, resolution and its elements
Open Designer	The drag and drop canvas, a button on each template

Print menu on records

Labels (designer)	On products, variants, transfers, lots, packages and locations
Labels button	In the header of every receipt, delivery and internal transfer

Printing (app)

Print Queue	Every job sent to a printer: queued, printed, failed, cancelled

Printing, Configuration

Printers	Each printer, what it prints and how Odoo reaches it
Printing Rules	Which report or label goes to which printer, and when
Label Templates	The same template list, when the Print Suite is installed
Workstations	Desks and stations people print from
Print Agents	The computers in the building that print for Odoo
Settings	Agent download, allowed server ports, how long jobs are kept

Your avatar, Preferences

Printing tab	My workstation, and Print without preview on or off

Who sees what

Each module has two access levels. They are given automatically to the people who already work in Inventory, and an administrator can change them on the user's form under the Labels and Direct Printing sections.

ACCESS LEVEL	MODULE	WHAT IT ALLOWS	GIVEN AUTOMATICALLY TO
User: print labels	LABEL DESIGNER	See templates, open the print wizard, print PDF and ZPL	Inventory users
Manager: design templates	LABEL DESIGNER	Create and change templates and save designs	Inventory administrators and Odoo administrators
User: print directly	DIRECT PRINT	See the Printing app and their own jobs; reports matching a rule print with no preview	Set by an administrator
Manager: printers, agents and rules	DIRECT PRINT	Configuration menu, test pages, pairing, every job of the company, Send again and Cancel	Odoo administrators

> In a company with several branches, every template, printer, agent, rule, workstation and job belongs to a company, and people only see those of the companies they work in. A label template or a workstation with no company is shared by all.

PART 3 Install and find your way in LABEL DESIGNER

Installing the Label Designer adds a Labels section to the Inventory configuration menu, six ready made templates, a Labels entry in the Print menu of six kinds of record, and a Labels button on every transfer.

IN THIS PART YOU WILL LEARN

- Installing the module and what arrives with it
- Where the templates live
- Where people print labels from

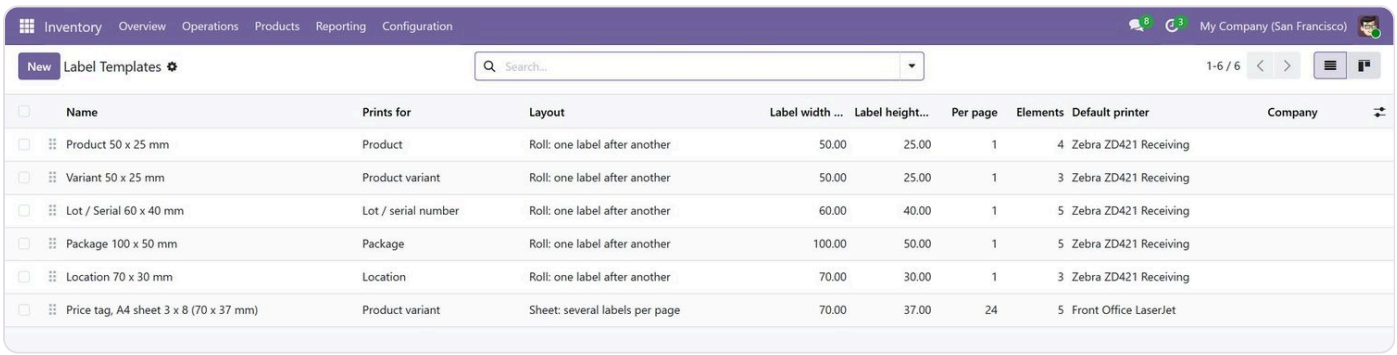
DO IT ON YOUR DATABASE

1. Open Apps, remove the Apps filter and search Label Designer, then Install
2. Open Inventory, Configuration, Label Templates
3. Open a receipt and look for the Labels button in its header

1 Open the template list

Go to **Inventory, Configuration, Labels, Label Templates**. The list shows each template's name, what it prints for, the layout, the width and height in millimetres, how many labels fit on a page, the number of elements and, with the Print Suite, its default printer.

> Drag the handle on the left to change the order. The first template of the right kind is the one the print wizard offers first.



The six templates that come with the module

TEMPLATE	PRINTS FOR	SIZE AND LAYOUT	WHAT IS ON IT
Product 50 × 25 mm	Product	50 × 25 mm roll	Product name, internal reference, sales price, product barcode
Variant 50 × 25 mm	Product variant	50 × 25 mm roll	Name with the variant, internal reference, product barcode
Lot / Serial 60 × 40 mm	Lot / serial number	60 × 40 mm roll	Product name, lot number, expiry date, a Data Matrix and a Code 128 of the lot
Package 100 × 50 mm	Package	100 × 50 mm roll	Company name, a rule line, package name, location, package barcode
Location 70 × 30 mm	Location	70 × 30 mm roll	A QR code of the location barcode, the location name and its code
Price tag, A4 sheet 3 × 8 (70 x 37 mm)	Product variant	A4 sheet, 3 across and 8 down	Company logo, name with the variant, price with currency, barcode, internal reference

They are ordinary templates: change them, copy them with **Action, Duplicate** (the copy is named with (copy) after it) or archive the ones you do not use.

Searching the list

SEARCH TOOL	WHAT IT DOES
Search by name	Type part of a template name
Roll / Sheet filters	Only roll templates, or only sheet templates
Archived filter	Shows templates you archived, so you can restore them
Group by Prints for / Layout	Groups the list by the kind of record, or by roll and sheet

The list also has a card view (the kanban icon at the top right) that shows each template's name, what it prints for, the size and the count per page, handy on a tablet.

PART 4 Set up a label template LABEL DESIGNER

A template is the label's paper: how big it is, whether labels come one after another on a roll or several to a sheet, and the resolution of the Zebra it is meant for. The elements on it are drawn in the designer in the next part.

IN THIS PART YOU WILL LEARN

- Every field on the template form and what it changes
- Roll and sheet layouts, columns, rows, margins and gaps
- What Prints for decides
- The checks Odoo makes on the size
- The Elements tab, for exact positions typed by hand

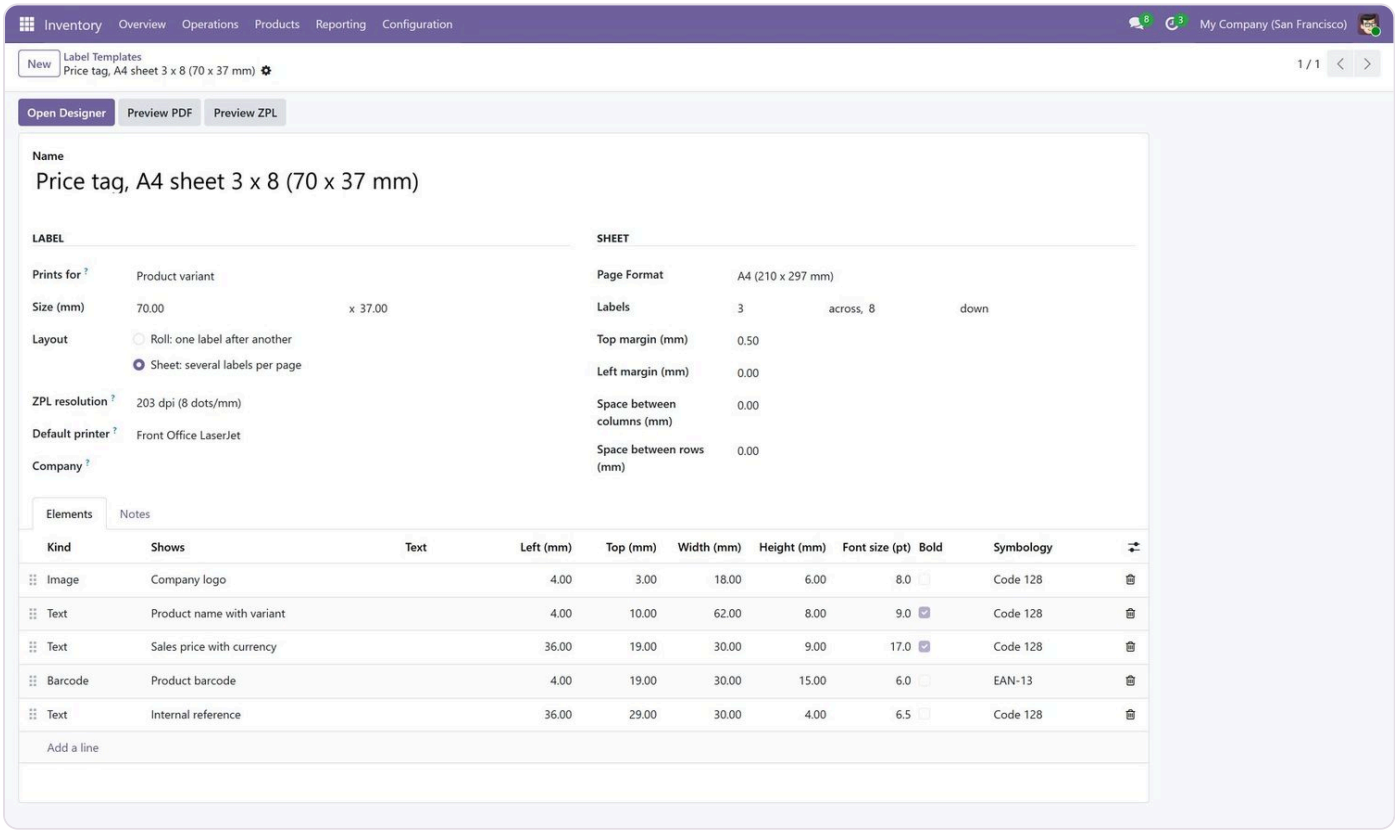
DO IT ON YOUR DATABASE

1. Open the Price tag, A4 sheet 3 × 8 template
2. Change the layout to Roll and see the Sheet group disappear, then Discard
3. Press Preview PDF to see it with a real product

1 Open a template

Click a template in the list. The header has three buttons: **Open Designer**, **Preview PDF** and **Preview ZPL**. Below the name, the **Label** group holds the size and the layout, and the **Sheet** group appears only for a sheet layout.

> Preview PDF and Preview ZPL open in a new tab and use the most complete record Odoo can find: for a product template, one that has both a barcode and an internal reference; for a lot, one with an expiry date; for a location, an internal one with a barcode.



The fields, one by one

FIELD	WHAT TO ENTER	NOTES
Name	What people will see in the print wizard, for example Shelf label 50 × 25	Can be translated
Prints for	Product, Product variant, Lot / serial number, Package or Location	Decides where the template is offered and which record a custom field path starts from

Size (mm)	Width x height of one label, in millimetres, with two decimals	A 2 inch roll label is 50.80 mm; decimals are kept exactly
Layout	Roll: one label after another. Sheet: several labels per page	A roll prints one label per PDF page, sized to the label
ZPL resolution	203 dpi (8 dots/mm), 300 dpi (12 dots/mm) or 600 dpi (24 dots/mm)	Must match the Zebra. A 203 dpi printer fed a 300 dpi label prints it two thirds too big
Default printer PRINT SUITE	The printer offered first when these labels are printed	A ZPL printer receives ZPL, a document printer the PDF. A receipt printer is refused
Company	Leave empty to share the template with every company	Only shown with several companies
Page format (sheet)	A4 (210 × 297 mm) or US Letter (215.9 x 279.4 mm)	
Labels (sheet)	Columns across and rows down	Labels per page is columns times rows
Top margin, Left margin (sheet)	Distance from the page edge to the first label	Measure it on your label stock with a ruler
Space between columns, rows (sheet)	The gap between labels	Zero for sheets where labels touch
Notes tab	Label stock, printer, anything the next person should know	Free text

The checks on the size

Odoo will not save a template that cannot print. The messages name the template and say exactly what is wrong:

WHEN	ODOO SAYS
Width or height is zero or less	a label needs a width and a height above zero
Width or height is over 1000 mm	a label larger than one metre is not a label
A margin or gap is negative	margins and spacing cannot be negative
A sheet with no columns or rows	a sheet needs at least one column and one row
The labels do not fit the page	names the labels, the space they need and the page size, for example 3 × 8 labels of 70 x 37 mm need 210.0 x 296.0 mm, the page is 210.0 x 297.0 mm

The Elements tab

Every element the designer places is also a line on the **Elements** tab. Here you can type positions exactly, which is quicker than dragging when you copy a layout from a supplier's spec sheet. The columns are Kind, Shows, Text, Left, Top, Width and Height in mm, Font size, Bold and Symbology; Field path, Alignment, Print the digits and Line thickness can be shown from the column picker on the right. Drag the handle to change which element is drawn on top.

ELEMENT CHECK	ODOO SAYS
Width or height zero or less	An element needs a width and a height above zero

Left or top below zero	An element cannot start outside the label
Text font outside 1 to 200 pt	A font size is between 1 and 200 points
Custom field with no path	A custom field element needs a field path
Logo or product image on a non-image element, or the other way round	The company logo and the product image go in an image element
A barcode with no symbology	Choose the barcode symbology

PART 5 Design the label in the designer LABEL DESIGNER

The designer is a canvas the size of your label, drawn to scale, with the real data of one of your records on it. You place text, barcodes, a logo, lines and boxes, move them with the mouse and set the rest in the panel on the right.

IN THIS PART YOU WILL LEARN

- The three columns of the designer
- The five kinds of element and the twenty things an element can show
- Moving, resizing and nudging elements with the keyboard
- Barcodes: the seven symbologies and the checks on the digits
- Custom field paths, for any field Odoo holds
- Saving, previewing and going back to the template

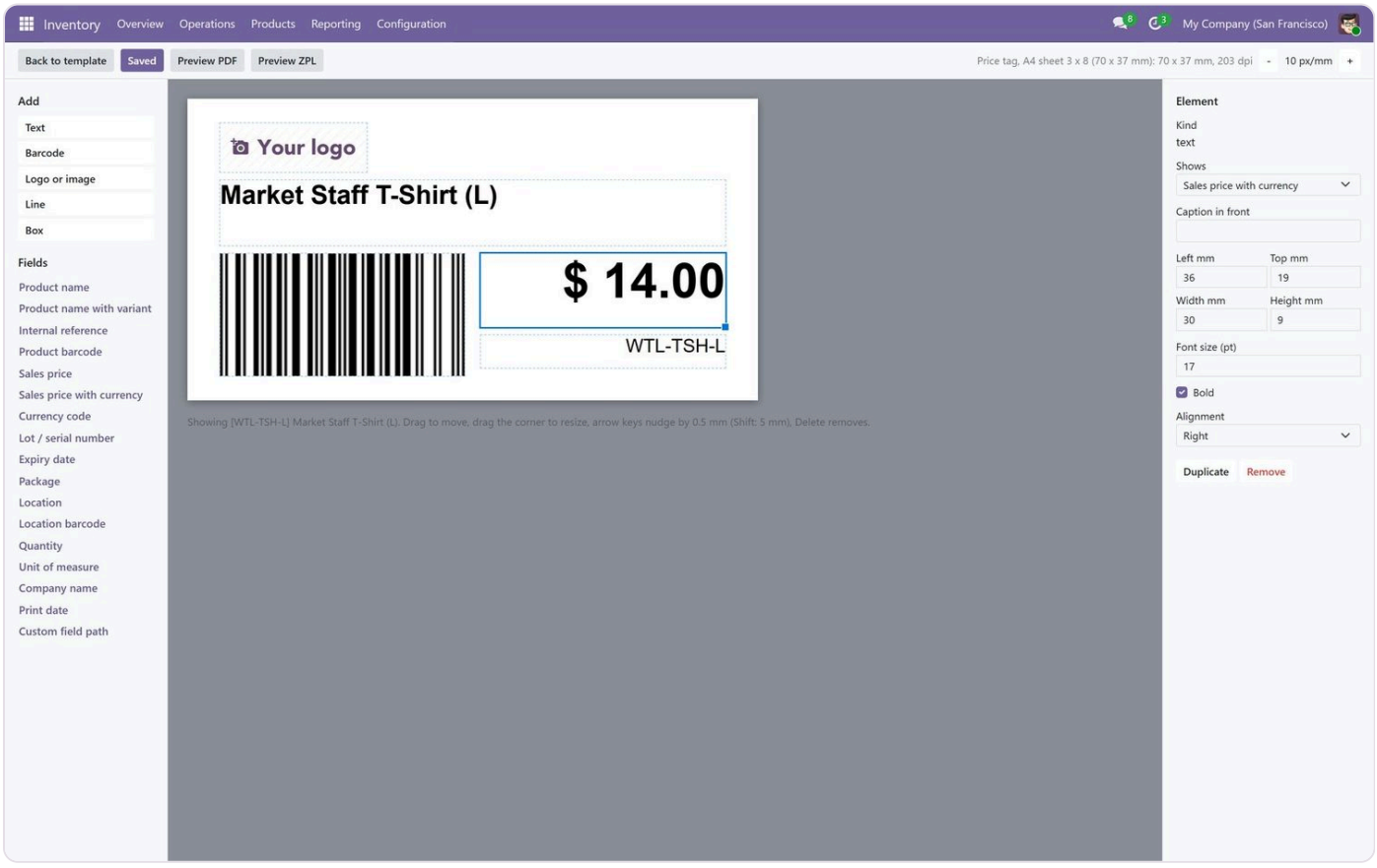
DO IT ON YOUR DATABASE

1. Open a template and press Open Designer
2. Click Barcode under Add, then pick Product barcode in Shows
3. Press Shift and an arrow key to move it 5 mm, then Save

1 Open the designer

Press **Open Designer** on the template. The left column holds **Add** (Text, Barcode, Logo or image, Line, Box) and **Fields** (one click adds a text element already showing that field). The middle is the label to scale. The right column is the selected element's settings. Across the top: **Back to template**, **Save**, **Preview PDF**, **Preview ZPL**, the template's name, size and dpi, and the zoom in pixels per millimetre with - and +.

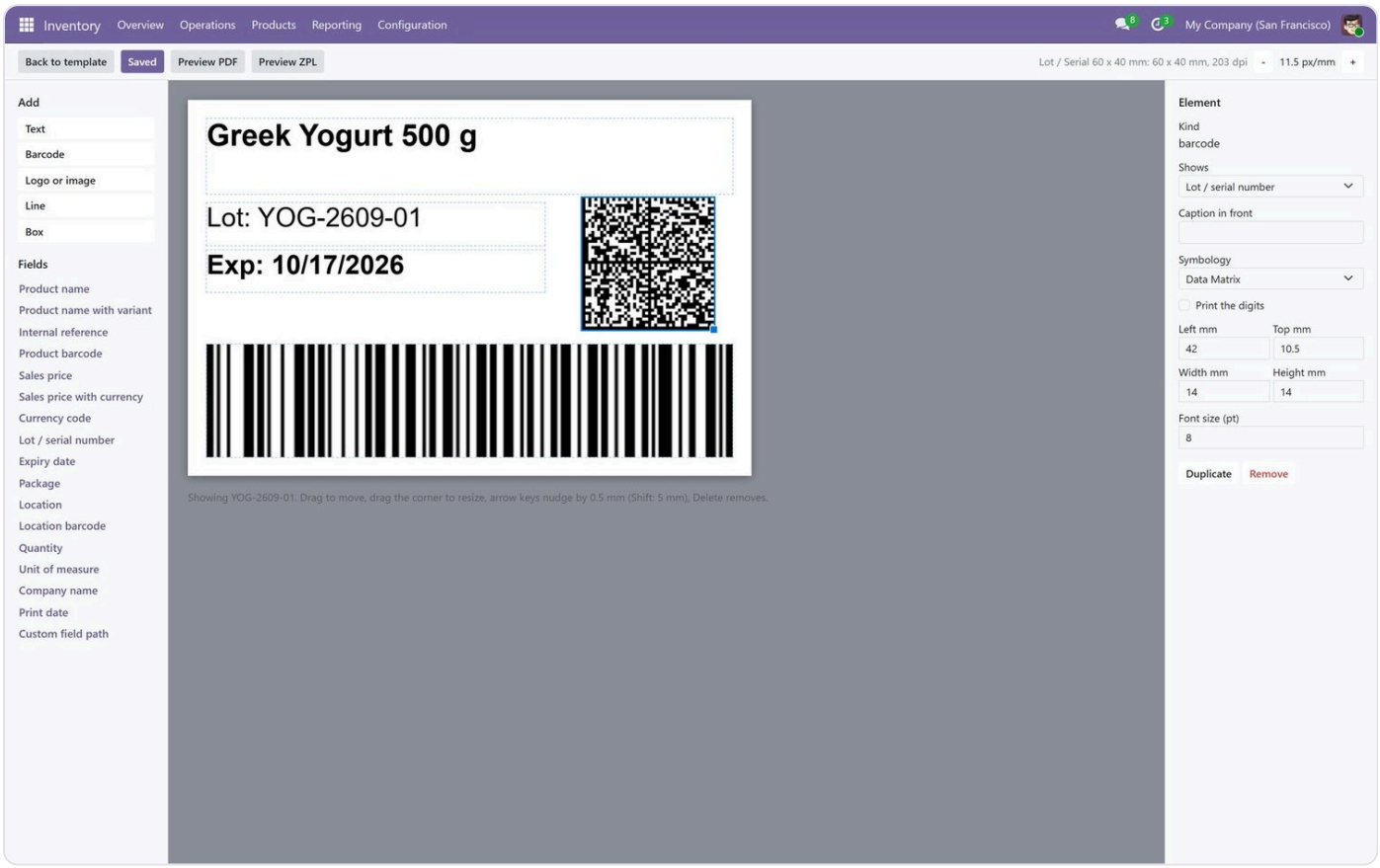
> Under the label the designer names the record it is showing, here a T-shirt variant, so you see your real names and prices rather than placeholders.



2 Select and change an element

Click an element to select it. The right panel shows its **Kind**, what it **Shows**, a **Caption in front** (for example Lot: or Exp:), **Left**, **Top**, **Width** and **Height** in mm, and, for text, **Font size**, **Bold** and **Alignment**. A barcode adds **Symbology** and **Print the digits**; a line or box adds the line thickness. **Duplicate** copies the element, **Remove** deletes it.

> Drag to move, drag the corner to resize, arrow keys nudge by 0.5 mm and with Shift by 5 mm, Delete removes the selected element. Click the grey area to select nothing.



The five kinds of element

KIND	USE IT FOR	SETTINGS
Text	Fixed words or the value of a field	Shows, caption, font size, bold, alignment left, centre or right
Barcode	A product, lot, package or location code as bars or a square code	Shows, symbology, print the digits
Image	The company logo or the product image	Shows (logo or image only)
Line	A rule across the label	Thickness in mm
Box	A frame around a block	Thickness in mm

What an element can show

SHOWS	WHAT PRINTS
Fixed text	The element's own text
Product name / Product name with variant	The name, or the name with the variant values

Internal reference / Product barcode	The product's reference and its barcode number
Sales price / Sales price with currency / Currency code	The price with the currency's decimals, the price formatted with the symbol, or the code such as USD
Lot / serial number / Expiry date	The lot, and its expiry date when the product has expiry dates (Odoo's Expiration Dates setting)
Package / Location / Location barcode	The package name, the full location name, and the location's barcode (the name when it has none)
Quantity / Unit of measure	The quantity of the transfer line and its unit
Company name / Company logo	Of the template's company, or the current one
Product image	The product photo
Print date	The date the label is printed
Custom field path	Any field, by a dotted path, explained below

Barcodes and the checks on them

SYMBOLOGY	CARRIES	WHAT THE DESIGNER CHECKS
EAN-13, EAN-8, UPC-A	Digits only, 13, 8 or 12 long	Digits only; one digit short gets its check digit added; a wrong check digit is refused and the right one named
Code 128	Any plain ASCII text	Other characters are refused with a suggestion to use a QR code
Code 39	Capitals, digits and - . space $ / + %	Lower case is printed as capitals; other characters are refused with a suggestion to use Code 128
QR code, Data Matrix	Any text, square	Always drawn square, in the smaller of width and height

A barcode that cannot be drawn shows the reason on the canvas instead of bars, so you see the problem while designing rather than at the printer. With **Print the digits** ticked, a line of digits is printed under the bars (not under QR and Data Matrix).

Custom field paths

Pick **Custom field path** and type a dotted path from the record the template prints for. For a product template, `categ_id.name` prints the product category; for a lot, `product_id.weight` prints the product's weight. Every step must be a real field and every step except the last must be a relation. Dates print in the user's date format, true and false print as Yes and No, a selection prints its label. A wrong path stops with a message naming the field that does not exist. Reading goes through Odoo's normal access rights, so a field the user may not read is not printed.

Saving

Save writes the design back to the template; the button reads **Saved** until you change something again. Elements you removed are deleted, new ones are created, and the order on the canvas becomes the order on the Elements tab. Only template managers can save; a user with print rights only can open the designer to look. **Back to template** returns to the form.

PART 6 Print labels LABEL DESIGNER

Labels are printed from the records they describe. Select one or many, open the Print menu and choose Labels (designer), or press the Labels button on a transfer. A wizard lists one line per label source with the number of labels each, and prints them all in one file.

IN THIS PART YOU WILL LEARN

- The six places labels are printed from
- One label per unit or per line on a transfer, and what per unit means for kilograms
- Labels each, Skip positions and the total
- Print PDF, Download ZPL and Show ZPL
- The limits and the messages you may meet

DO IT ON YOUR DATABASE

1. Open a validated receipt and press Labels
2. Switch How many between One label per unit and One label per line and watch Labels in total
3. Press Show ZPL, then Cancel

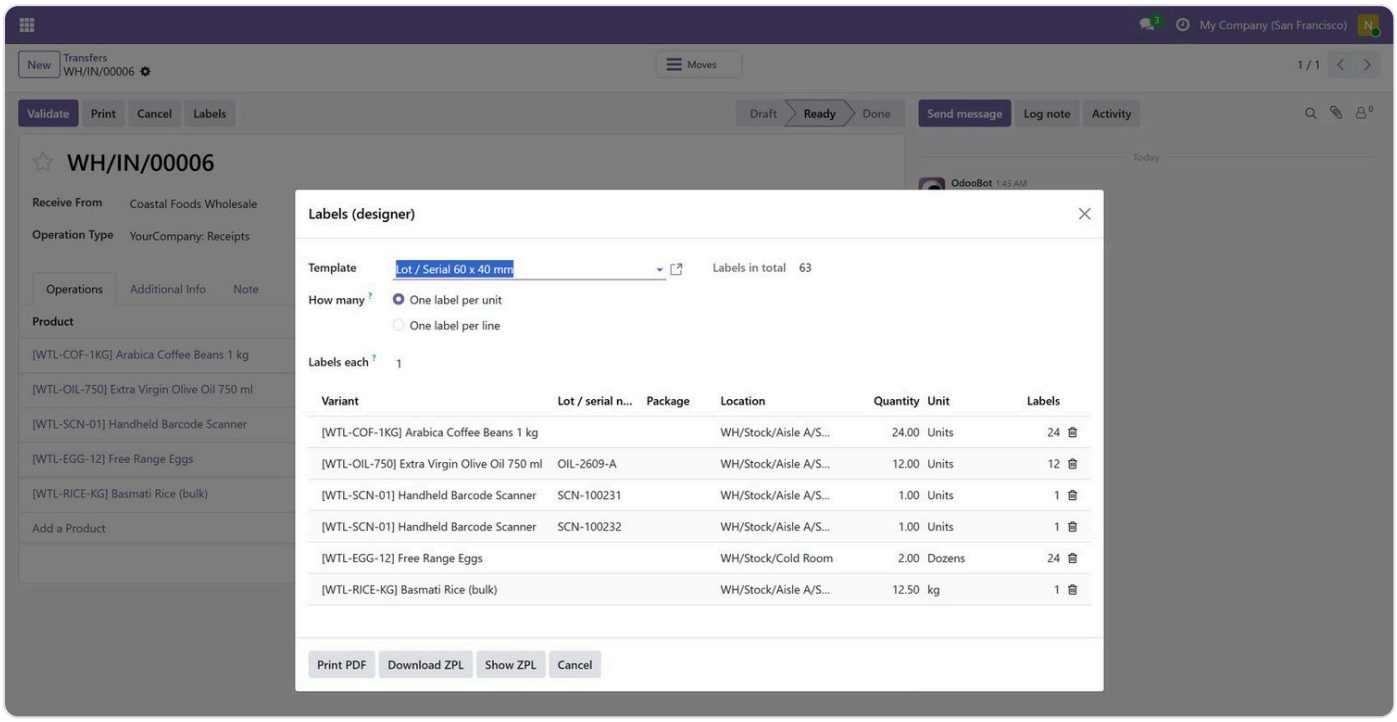
Where labels are printed from

RECORDS	WHERE	TEMPLATES OFFERED
Products	Inventory or Sales, Products: list, card or form, Print menu	Product and Product variant templates
Product variants	The variants list or form, Print menu	Variant and Product templates
Transfers (receipts, deliveries, internal)	Print menu on the list or form, or the **Labels** button in the transfer header (not on cancelled transfers)	Variant, Product, Lot and Package templates
Lots / serial numbers	Inventory, Products, Lots / Serial Numbers, Print menu	Lot templates
Packages	Inventory, Products, Packages: list, card or form, Print menu	Package templates
Locations	Inventory, Configuration, Locations, Print menu	Location templates

1 Open the wizard from a receipt

On a receipt, press **Labels**. The wizard opens with one line for each operation line that has a quantity: the variant, the lot or serial number, the package, the destination location, the quantity, its unit and the number of labels. When any line carries a lot, a lot template is chosen first; otherwise the first roll template of the right kind.

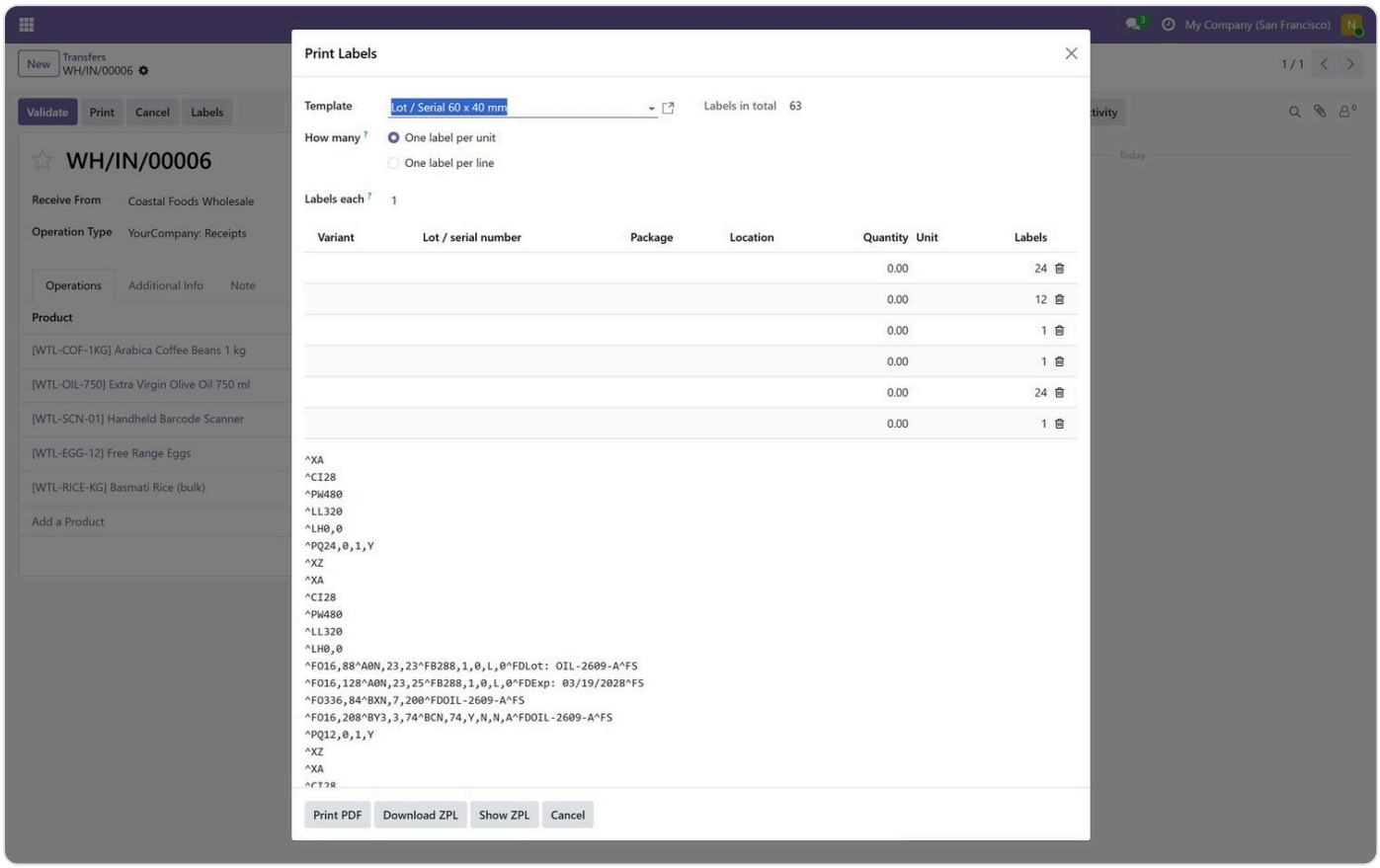
> A transfer with no quantities yet stops with: These transfers have no quantities yet. Set the quantities (or check availability) before printing labels for them.



2 Choose the counts, then print

Template lists only templates that suit the records. **How many** (transfers only): **One label per unit** prints 12 labels for 12 units, rounded up, and counts dozens as twelve units; a quantity in kilograms, litres or metres prints one label for the line. **One label per line** prints one for each line. **Labels each** multiplies the rule; elsewhere it is the number per record. You can still type the count of any line by hand. **Labels in total** adds them up. Here **Show ZPL** has been pressed: the ZPL appears under the lines.

> On a sheet template, **Skip positions** leaves that many places empty at the start of the first page, so a sheet that was half used last week can go back in the printer.



The buttons at the bottom

BUTTON	WHAT HAPPENS
Print PDF	Opens the labels as one PDF in a new tab, page size taken from the template
Download ZPL	Downloads a .zpl file to send to a Zebra, for example by copying it to the printer's share
Show ZPL	Shows the ZPL text in the wizard, for checking or pasting into a viewer
Send to printer PRINT SUITE	Queues the labels for the chosen printer, explained in part 14
Cancel	Closes the wizard, nothing is printed

Limits and messages

WHEN	ODOO SAYS
Every count is zero	There is nothing to print: every label count is zero
More than 5,000 labels in one print	names the total and asks to lower the counts or print in parts. A slip of the finger on a receipt of 90,000 screws does not empty the label roll
A negative count	A label count cannot be negative

The template has no elements	The template has no elements yet. Open it in the designer first
The template is for another kind of record	The template is not for these records
Nothing selected	Select the records to print labels for first

> The PDF and ZPL links only open for the person who opened the wizard. A colleague who guesses the address gets nothing.

PART 7 One design, two outputs: PDF and ZPL LABEL DESIGNER

Every template prints both ways from the same design. PDF goes to any printer, at true size. ZPL is the language Zebra and compatible thermal printers read; it prints faster, sharper and without a driver.

IN THIS PART YOU WILL LEARN

- How the PDF is sized, for rolls and for sheets
- What the ZPL contains and why the dpi matters
- How to check a ZPL label before you own a Zebra



PDF. The lot label from the Lot / Serial 60 × 40 mm template, product name, lot, expiry, Data Matrix and Code 128. On a roll, each label is its own page of exactly 60 × 40 mm.



ZPL. The same label from the same template, drawn from its ZPL by a ZPL viewer. Positions, fonts and barcodes are computed in printer dots for the template's dpi.

The PDF

A roll template prints one label per page with the page set to the label's size, to a tenth of a millimetre, with no margins, so the printer driver prints at 100 percent and nothing is shrunk. A sheet template prints on A4 or Letter, filling positions left to right and top to bottom, with your margins and gaps. Barcodes are drawn at 300 dpi, which is sharp on a laser and far beyond a thermal printer.

The ZPL

Each label is one ZPL block; identical copies are printed by the printer itself with a print quantity, so 24 copies of a label cost the network one label's worth of data. Millimetres become dots for the template's resolution (8, 12 or 24 dots per mm), text uses the printer's scalable font, and EAN, UPC, Code 128, Code 39, QR and Data Matrix are sent as the printer's own barcode commands, so they are drawn by the printer at full sharpness. Images such as a logo are sent as a graphic.

> Before you buy a printer, paste the text from **Show ZPL** into an online ZPL viewer such as labelary.com to see exactly what the Zebra will print. Set the viewer to the same dpi and label size as the template.

Before you go live with labels

CHECK	WHY

Measure the label stock and set width, height, margins and gaps from it	Labels that are 1 mm off drift across a sheet
Set the ZPL resolution to the printer's	A wrong dpi prints too big or too small
Print one PDF at 100 percent (no "fit to page")	Driver scaling is the usual cause of labels that do not line up
Scan one printed barcode with your scanner	Proves the size and the quiet zones are right
Give products barcodes and internal references	Empty fields print nothing; the element is simply left out

PART 8 How Direct Print works DIRECT PRINT

Direct Print changes one thing people see: when a report has a printing rule, clicking Print sends it to a printer and shows a short notice instead of opening a PDF. Everything else is the set up behind that click.

IN THIS PART YOU WILL LEARN

- The four things you set up: agents, printers, workstations and rules
- The two ways a job reaches a printer
- What happens when nothing matches

What happens when someone presses Print

1. **Print** — Invoice, slip, any PDF or text report
2. **Rule lookup** — Report, user, workstation
3. **Render** — As that user, with their rights
4. **Print queue** — A job, numbered PJ/00001
5. **Printer** — By the agent or the server

The four building blocks

BLOCK	WHAT IT IS	PART
Print agent	A small program on a computer in the building that fetches jobs over HTTPS and prints them	9 and 10
Printer	One physical printer: what language it speaks and how Odoo reaches it	11
Workstation	A desk or station people print from, such as the receiving desk	12
Printing rule	Which report, for whom, goes to which printer, how many copies, and when	13

Two ways to reach a printer

CONNECTION	USE IT WHEN	HOW
Through a print agent in the building	Odoo runs in the cloud, or the printer is a USB printer on a PC, or it is an office printer that needs its driver	The agent asks Odoo for work every few seconds over HTTPS. Nothing has to be opened on the firewall
Straight from the Odoo server	Odoo runs on the same network as a network label or receipt printer	The server opens a connection to the printer's IP address and port, only on ports you allow

If no rule matches, the user has switched **Print without preview** off, or the user does not have the Direct Printing user right, the report downloads exactly as it does in standard Odoo. Nothing breaks for people the rules do not cover.

PART 9 Settings and the print agent DIRECT PRINT

Start in Settings. It holds the agent download and the two server settings. Then install the agent on the computer next to your printers.

IN THIS PART YOU WILL LEARN

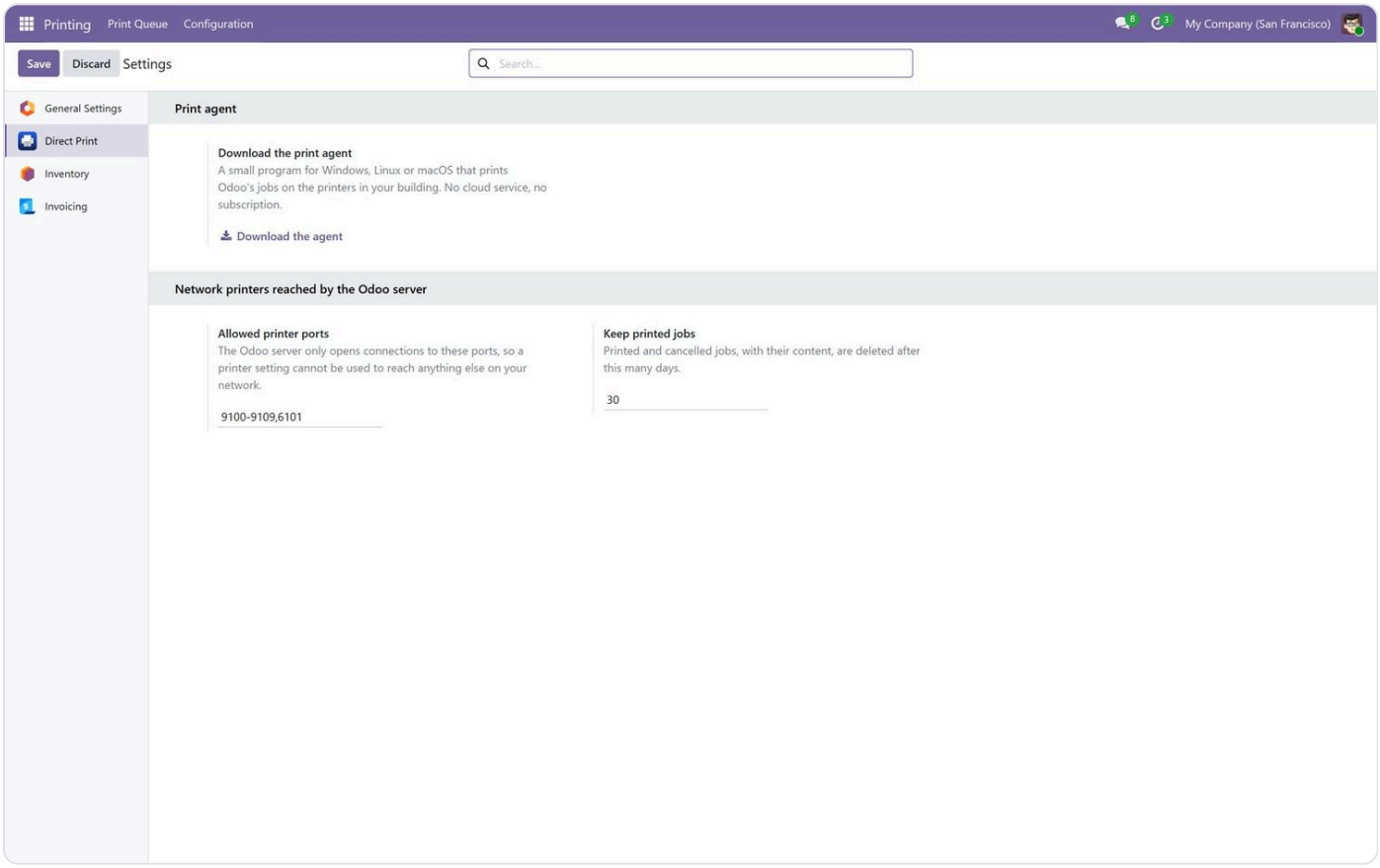
- Downloading the agent
- Installing it on Windows, Linux and macOS
- The everyday agent commands and where its log is
- Allowed printer ports and how long jobs are kept

DO IT ON YOUR DATABASE

1. Open Printing, Configuration, Settings
2. Press Download the agent and unzip it on the printing computer
3. Run the install steps for that system

1 Open the Direct Print settings

Go to **Printing, Configuration, Settings** (it opens the Direct Print section of Odoo's settings). Only print managers see it.



SETTING	WHAT IT DOES	DEFAULT
Download the agent	Downloads the agent as a zip, built from the copy inside the module, so it always speaks the same protocol as your Odoo	
Allowed printer ports	The only ports the Odoo server may open to a network printer, as ranges and single ports separated by commas	9100-9109,6101
Keep printed jobs	Printed and cancelled jobs, with their content, are deleted after this many days (at least one)	30 days

Install the agent on Windows

Install Python 3.10 or newer from python.org and tick Add python.exe to PATH. Unzip the agent to `C:\Program Files\WT Print Agent`, open PowerShell as administrator in that folder and install what it needs:

```
python -m pip install -r requirements.txt
```

Pair it (part 10 shows where the code comes from), then make it start with Windows. The first line runs it for the signed in user, who sees every printer that user sees, shared office printers included; the second runs it at boot, before anyone signs in, with the printers installed for all users:

```
powershell -ExecutionPolicy Bypass -File install-windows.ps1
powershell -ExecutionPolicy Bypass -File install-windows.ps1 -AtStartup
```

Install on Linux, or a Raspberry Pi next to a Zebra

```
sudo apt install python3-requests cups-client
sudo mkdir -p /opt/wt-print-agent && sudo cp -r . /opt/wt-print-agent
cd /opt/wt-print-agent
sudo python3 -m wt_print_agent pair --url https://your.odoo.com --db yourdb --code WTP-XXXX-XXXX-XXXX
sudo cp wt-print-agent.service /etc/systemd/system/
sudo systemctl enable --now wt-print-agent
```

On macOS the same commands work without systemd; run `python3 -m wt_print_agent run` from a LaunchAgent.

Everyday agent commands

COMMAND	WHAT IT DOES
`python -m wt_print_agent printers`	Lists the printers this computer has
`python -m wt_print_agent status`	Shows where the agent points
`python -m wt_print_agent test-socket --host 192.168.1.50`	Sends a ZPL test label to a network printer
`python -m wt_print_agent run --verbose`	Runs in the foreground so you can watch it work
`python -m wt_print_agent version`	Prints the agent version

The agent prints PDF documents through the printer's own driver at 100 percent (on Windows it also sets the paper size from the PDF page), ZPL and ESC/POS as raw bytes to a printer installed on the computer, and ZPL and ESC/POS straight to a network printer on port 9100. Its log, `agent.log`, and its settings are in `C:\ProgramData\WTPrintAgent\` on Windows and in `/etc/wt-print-agent/` or `~/.config/wt-print-agent/` on Linux. The token is stored there, encrypted for the machine on Windows and readable by its owner only on Linux.

> Testing without paper: in the agent's `config.json`, map a printer name to a folder under `file_printers`, and jobs for that printer are written to the folder as PDF files.

PART 10 Print agents and pairing DIRECT PRINT

An agent record in Odoo stands for one computer running the agent. Pairing gives that computer its own secret token, once, with a one time code.

IN THIS PART YOU WILL LEARN

- Creating the agent record
- Pairing with a one time code, and pairing again
- Revoking a lost or rebuilt computer
- What the agent reports back: computer, system, version and its printers

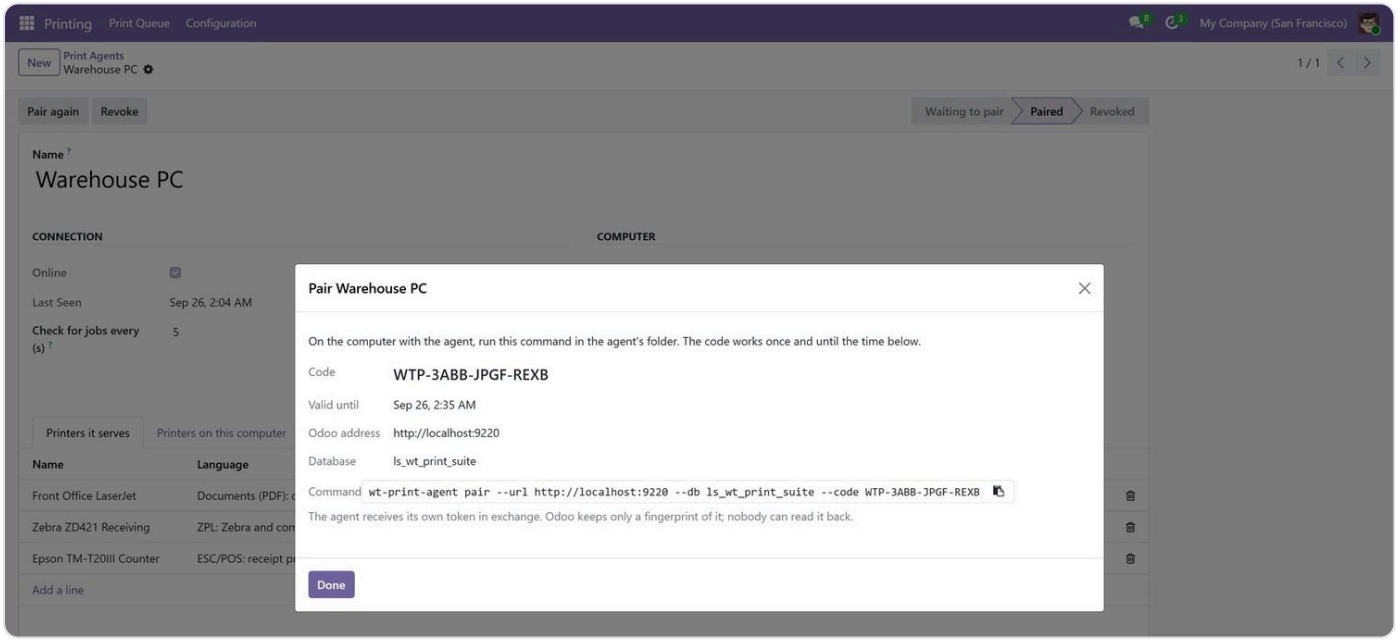
DO IT ON YOUR DATABASE

1. Open Printing, Configuration, Print Agents and create Warehouse PC
2. Press Pair, copy the command and run it in the agent's folder
3. Reload the form: Online is ticked and the Printers on this computer tab is filled

1 Create the agent and press Pair

In **Printing, Configuration, Print Agents**, press **New**, name it after the computer (Warehouse PC, Front office) and press **Pair**. A dialog shows a code such as WTP-3ABB-JPGF-REXB, the time it is valid until (30 minutes), the Odoo address, the database and the full command with a copy button. Run it in the agent's folder, then press **Done**.

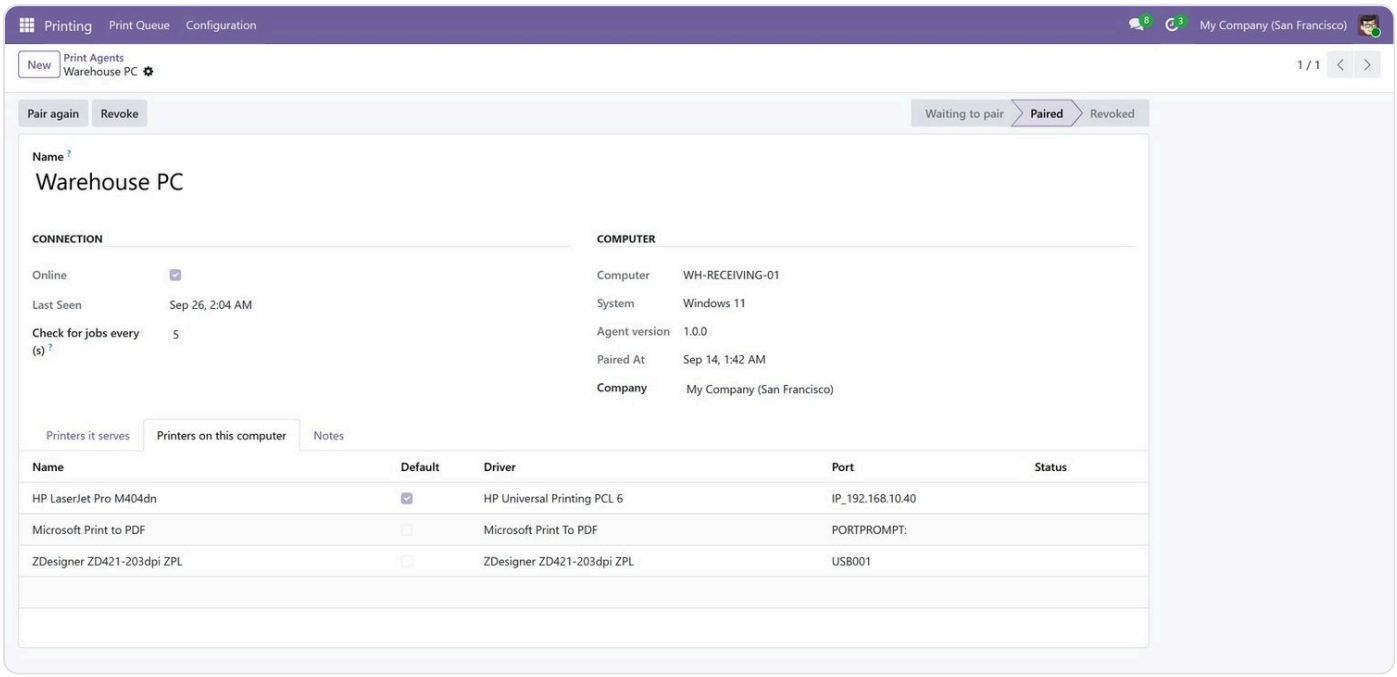
> In the folder you unzipped, type the command as `python -m wt_print_agent pair --url ... --db ... --code ...`, with the three values from the dialog. The code works once. Odoo keeps only a fingerprint of the token the agent receives; nobody can read it back.



2 Check the agent is online

Once paired, the status bar moves from **Waiting to pair** to **Paired**. The **Connection** group shows **Online**, **Last seen** and **Check for jobs every (s)**, which you can set from 2 to 60 seconds (5 by default). The **Computer** group shows the computer name, its system, the agent version and when it was paired. The tab **Printers on this computer** lists what the operating system reports: name, default, driver, port and status.

> An agent counts as online when it checked in within the last few polls. A printer served by an offline agent shows Agent offline, and its jobs wait in the queue until the agent is back.



BUTTON OR TAB	WHAT IT DOES
Pair	Shown while the agent waits to pair: makes a new code
Pair again	Shown on a paired agent, asks to confirm. The current token stops working until the new code is used. This is also how you replace a stolen or rebuilt computer
Revoke	Asks to confirm, then the agent stops printing at once and has to be paired again
Printers it serves	The Odoo printers that go through this agent. You can add one here
Notes	Free text

PART 11 Printers DIRECT PRINT

A printer record says what a printer prints and how Odoo reaches it. Set each one up once, print a test page, and it is ready for rules.

IN THIS PART YOU WILL LEARN

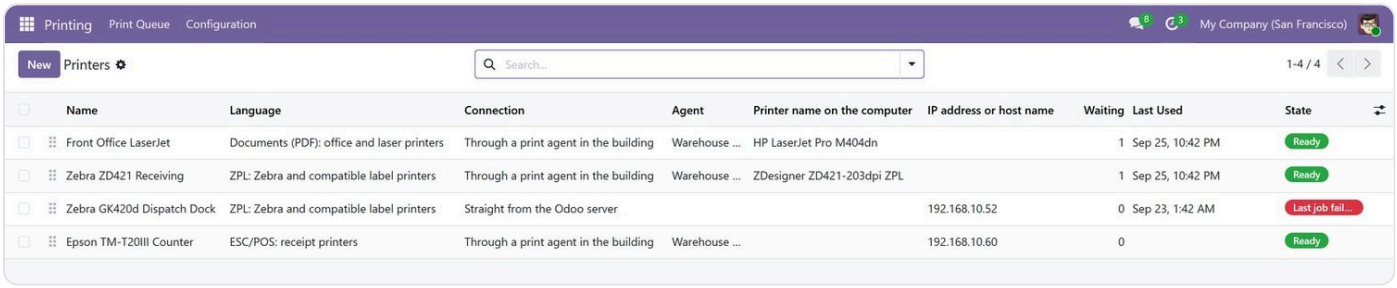
- The four printer languages and what each accepts
- Through an agent or straight from the server, by name or by IP
- The test page, the status and the last problem

DO IT ON YOUR DATABASE

1. Open Printing, Configuration, Printers and create a printer
2. Pick its language and connection, then pick it from the agent
3. Press Print a test page and look in the Print Queue

1 The printer list

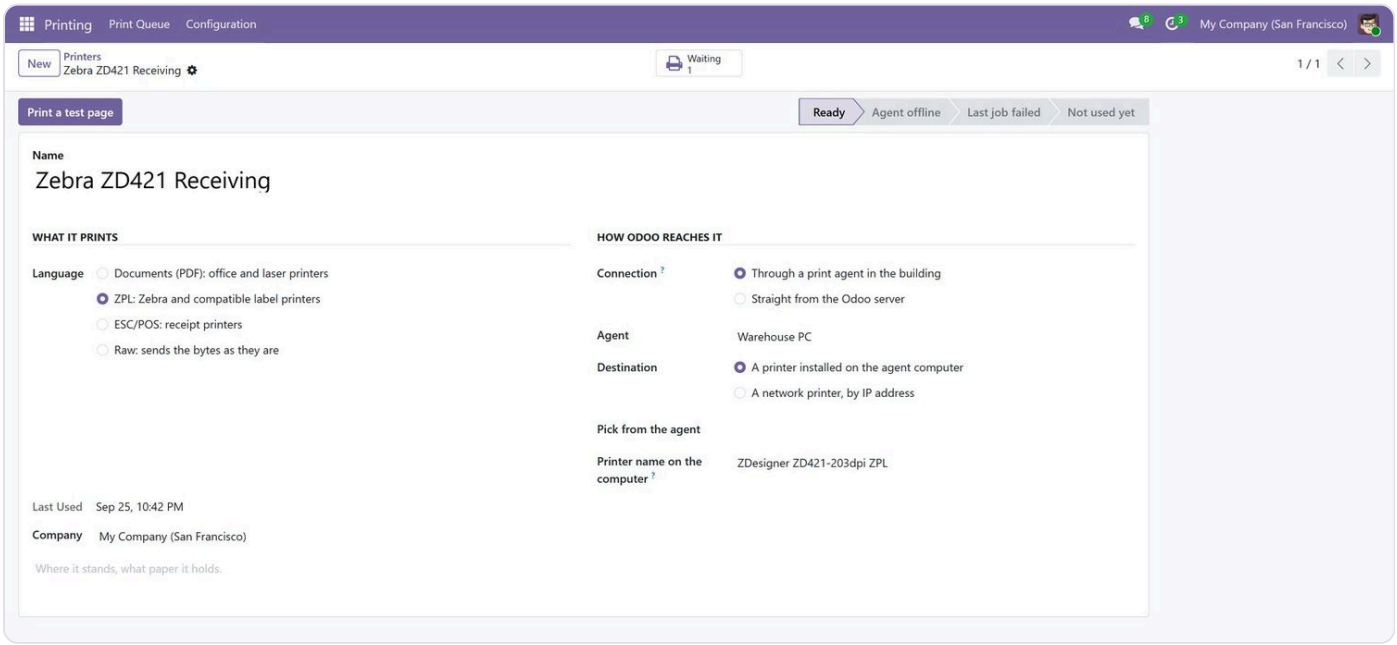
Printing, Configuration, Printers lists each printer with its language, connection, agent, the name on the computer, the IP address, the jobs waiting, when it was last used and a status badge: **Ready**, **Agent offline**, **Last job failed** or **Not used yet**. Filters pick label printers or document printers; group by connection or language.



2 A printer form

What it prints is the language. **How Odoo reaches it** is the connection; through an agent you choose the **Agent** and the **Destination**: a printer installed on the agent computer (pick it from the agent's list, which fills **Printer name on the computer**) or a network printer by IP address and port. The **Waiting** button opens its jobs. **Print a test page** queues a test for this printer.

> Type the name exactly as the computer shows it, for example ZDesigner ZD421-203dpi ZPL. Picking it from the agent avoids typing errors.



Languages

LANGUAGE	FOR	ACCEPTS
Documents (PDF)	Office and laser printers	PDF reports and PDF labels
ZPL	Zebra and compatible label printers	ZPL and raw text reports
ESC/POS	Receipt printers	ESC/POS and raw text reports
Raw	Anything else: the bytes go as they are	Raw, ZPL and ESC/POS

The checks on a printer

WHEN	ODOO SAYS
Agent connection with no agent	choose the print agent that serves this printer
Server connection to a printer by name	the Odoo server can only reach a network printer by its IP address
Installed printer with no name	type the printer name as the computer shows it, or pick it from the agent
A server printer on a port that is not allowed	names the port; add it under Settings, Allowed printer ports
PDF to a network port	most printers print a PDF sent to their network port as pages of code. Print it through an agent, or tick **Printer reads PDF directly** once a test page proved it does

The test page is a one page PDF for a document printer, a small ZPL label for a label printer, and a short ticket with a paper cut for a receipt printer, each naming the printer, the company and the time. When a job fails, the reason shows in red under **Last problem** on the form. Moving a printer to another agent moves its waiting jobs with it.

PART 12 Workstations and personal preferences DIRECT PRINT

A printer is a physical thing next to a desk. A workstation names that desk, so a rule can say 'invoices printed at the front office go to the front office LaserJet', whoever is sitting there.

IN THIS PART YOU WILL LEARN

- Creating workstations
- How people choose where they are
- Switching printing without preview off for yourself

DO IT ON YOUR DATABASE

1. Open Printing, Configuration, Workstations and add Receiving desk
2. Open your avatar, Preferences, the Printing tab
3. Set My workstation and keep Print without preview ticked

FIELD	ON	WHAT IT DOES
Name, Code	Workstation	The desk, counter or packing station, and an optional short code
People here now	Workstation	The users who chose this workstation, read only
Rules	Workstation form	The printing rules for this workstation; add one here and it is filled in
Company	Workstation	Empty means shared by every company
My workstation	Preferences, Printing	Where you work now. Only workstations of your companies are accepted
Print without preview	Preferences, Printing	Ticked by default. Untick it and your reports download as usual, even when a rule matches

The workstation list can be edited in place: type a name and a code on a new line. An administrator can set both preferences for anyone on the user's form, under its **Printing** tab.

> Each person sets their workstation from their own avatar menu, **Preferences**, tab **Printing**. A warehouse clerk who moves from receiving to packing changes it there, and their labels follow them to the next printer.

PART 13 Printing rules DIRECT PRINT

A rule names a printer and, if you want, narrows itself to one report, one user and one workstation, with a number of copies. With the free companions a rule can also fire on its own when a transfer is validated or an invoice is confirmed.

IN THIS PART YOU WILL LEARN

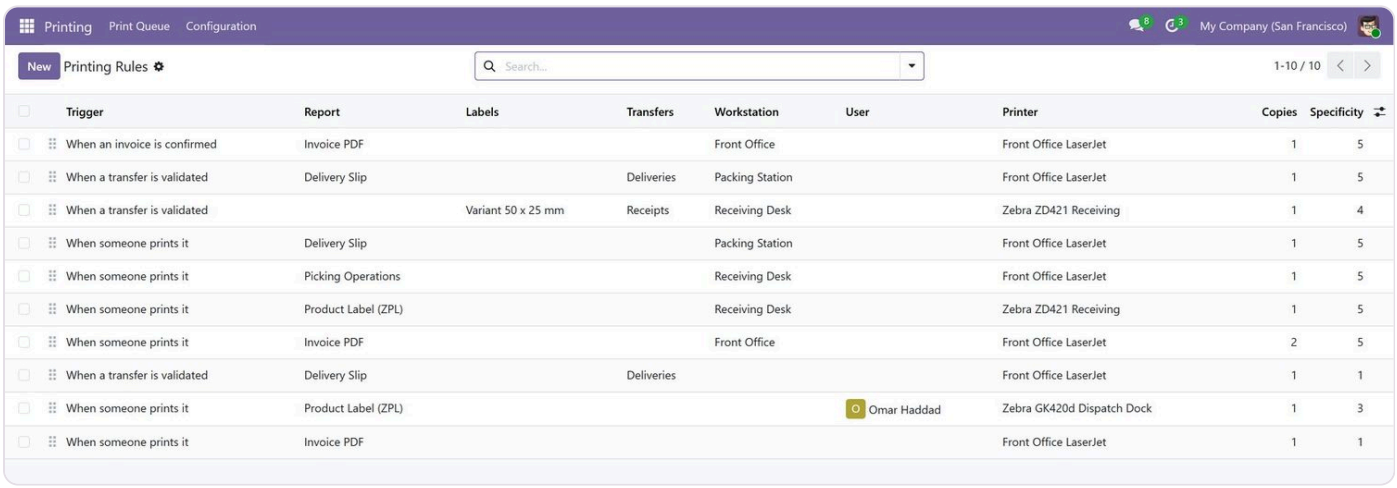
- Writing a rule for a report
- Which rule wins when several match
- Automatic printing on transfers and on invoices
- The checks Odoo makes on a rule

DO IT ON YOUR DATABASE

1. Open Printing, Configuration, Printing Rules and create one
2. Choose Invoice PDF, the Front office workstation and the front office printer
3. Print an invoice from that workstation: a green notice names the job and the printer

1 The rule list

Printing, Configuration, Printing Rules lists every rule with its trigger, report, the kind of transfer, labels (with the Print Suite), workstation, user, printer, copies and specificity. Drag the handle to change the sequence. Group by When or by Printer.

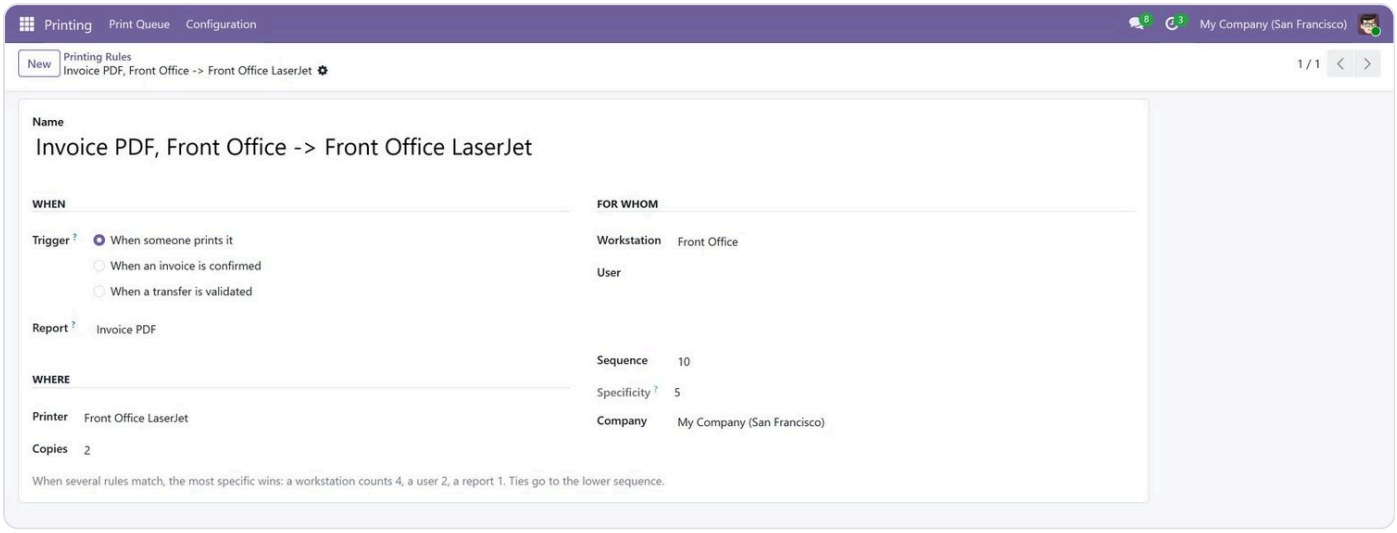


Trigger	Report	Labels	Transfers	Workstation	User	Printer	Copies	Specificity
When an invoice is confirmed	Invoice PDF			Front Office		Front Office LaserJet	1	5
When a transfer is validated	Delivery Slip		Deliveries	Packing Station		Front Office LaserJet	1	5
When a transfer is validated		Variant 50 x 25 mm	Receipts	Receiving Desk		Zebra ZD421 Receiving	1	4
When someone prints it	Delivery Slip			Packing Station		Front Office LaserJet	1	5
When someone prints it	Picking Operations			Receiving Desk		Front Office LaserJet	1	5
When someone prints it	Product Label (ZPL)			Receiving Desk		Zebra ZD421 Receiving	1	5
When someone prints it	Invoice PDF			Front Office		Front Office LaserJet	2	5
When a transfer is validated	Delivery Slip		Deliveries			Front Office LaserJet	1	1
When someone prints it	Product Label (ZPL)				Omar Haddad	Zebra GK420d Dispatch Dock	1	3
When someone prints it	Invoice PDF					Front Office LaserJet	1	1

2 A rule form

When: the trigger and the report. **For whom**: a workstation and a user, both optional. **Where**: the printer and the copies. The name fills itself, for example Invoice PDF, Front Office → Front Office LaserJet, and you can change it.

> Leave the report empty for a catch all rule that sends every report a person prints to one printer.



Which rule wins

When several rules match, the most specific one wins. A workstation counts 4, a user 2 and a report 1; the sum is the rule's **Specificity**. Ties go to the lower sequence.

RULE	SPECIFICITY
Invoices at the front office	4 + 1 = 5
Everything Sara prints	2
All invoices	1
Catch all	0

A rule only matches a printer that can take what the report produces: a PDF report needs a document printer; a text report (a ZPL or ESC/POS template) goes to a printer of that language. Archived printers are skipped.

The triggers

WHEN	MODULE	EXTRA FIELDS
When someone prints it	Direct Print	None
When a transfer is validated	Direct Print for Inventory (free)	**Transfers**: Receipts, Deliveries or Internal transfers, empty for all. **Operation types**: only these, empty for all of that kind. The report must be one of the transfer, such as Delivery Slip
When an invoice is confirmed	Direct Print for Invoicing (free)	**Invoices**: customer invoices and credit notes (default), vendor bills and refunds, or every invoice and bill. **Journals**: empty for every journal. The report must be one of the invoice

An automatic rule prints after the business action succeeded and never undoes it. The document is rendered as the person who validated or confirmed it, so what they may not read, they do not print. If printing fails, the transfer or invoice gets a note in its chatter, Automatic printing on (printer) failed, and the rest carry on. Per kind of document, only the most specific automatic rule for that person and workstation prints.

The checks on a rule

WHEN	ODOO SAYS
Printer of another company	the printer belongs to another company
Workstation of another company	the workstation belongs to another company
Automatic rule with nothing to print	an automatic rule needs the document to print
Report the printer cannot print	names the report, its kind (PDF) and the printer
Transfer trigger with a report of another model	the report must be one of the transfer
Invoice trigger with a report of another model	the report must be one of the invoice

PART 14 Labels straight to the printer PRINT SUITE

With both paid modules installed, the free Print Suite installs itself and joins them up: labels go to a printer from the print wizard, each template can name its printer, and receipts print their labels the moment they are validated.

IN THIS PART YOU WILL LEARN

- A default printer on a template
- Send to printer in the print wizard
- A rule that prints labels when a receipt is validated

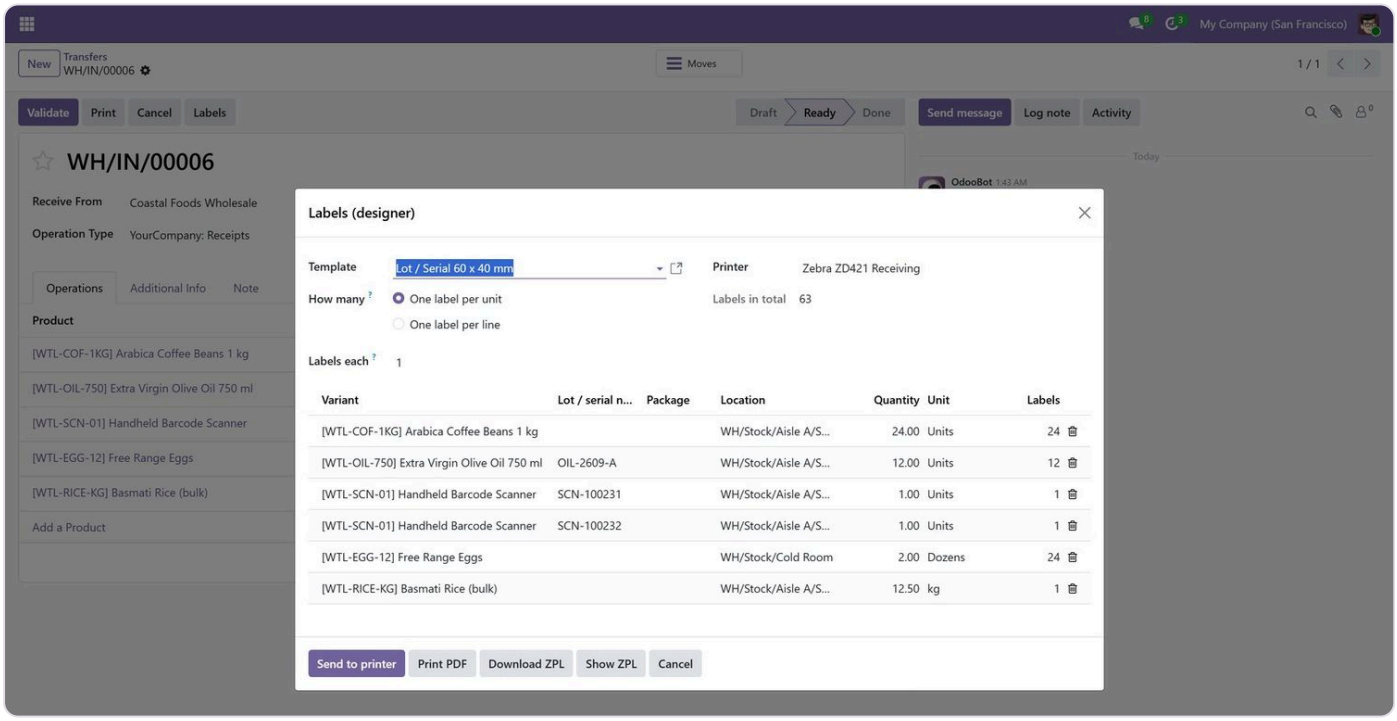
DO IT ON YOUR DATABASE

1. Open the Lot / Serial template and set Default printer to your Zebra
2. Open a receipt, press Labels, check the printer and press Send to printer
3. Write a label rule for receipts and validate the next receipt

1 Send labels to the printer

In the print wizard, a **Printer** field appears for people with the Direct Printing user right, filled from the template's **Default printer**. With a printer chosen, **Send to printer** becomes the main button. A ZPL printer receives ZPL made from the design; a document printer receives the PDF. A notice confirms, for example PJ/00011: 30 labels sent to Zebra ZD421 Receiving.

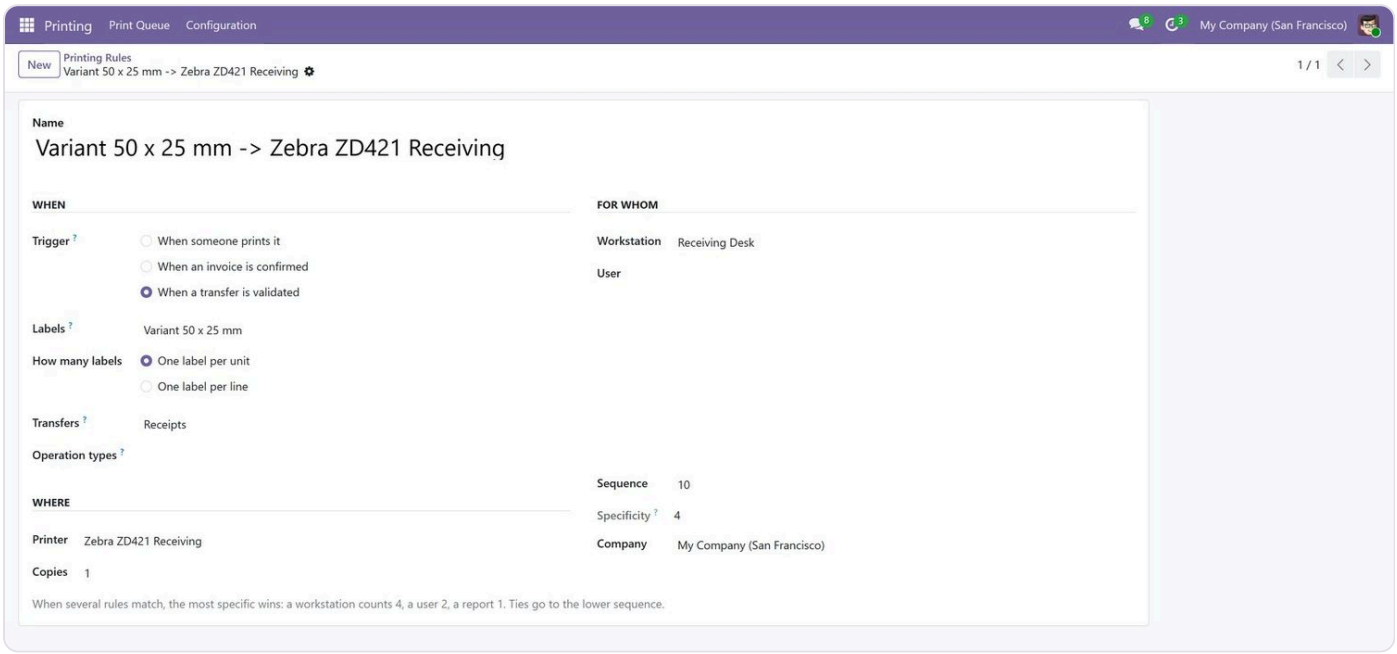
> Only document, ZPL and raw printers are offered. A receipt printer cannot print labels.



2 Labels printed when a receipt is validated

Create a rule with **When a transfer is validated**, leave the report empty and choose a **Labels** template (variant, product, lot or package) and **How many labels**: one per unit or one per line. Narrow it to Receipts. The name fills itself, for example Variant 50 × 25 mm → Zebra ZD421 Receiving. From then on, validating a receipt puts one job in the queue with every line's labels.

> A rule prints either a report or labels, not both; labels need the transfer trigger and a printer that can print labels. Odoo refuses anything else with a message saying which.



The **Label Templates** list also appears under **Printing, Configuration**, and shows each template's default printer as a column.

PART 15 The print queue DIRECT PRINT

Every page Odoo sends to a printer is a job here first, whichever way it travels. It is the one place to see what printed, what did not and why, and to send it again.

IN THIS PART YOU WILL LEARN

- The five states of a job
- Retries, and what happens when an agent goes quiet
- Send again, Cancel and Download
- Filters, and how long jobs are kept

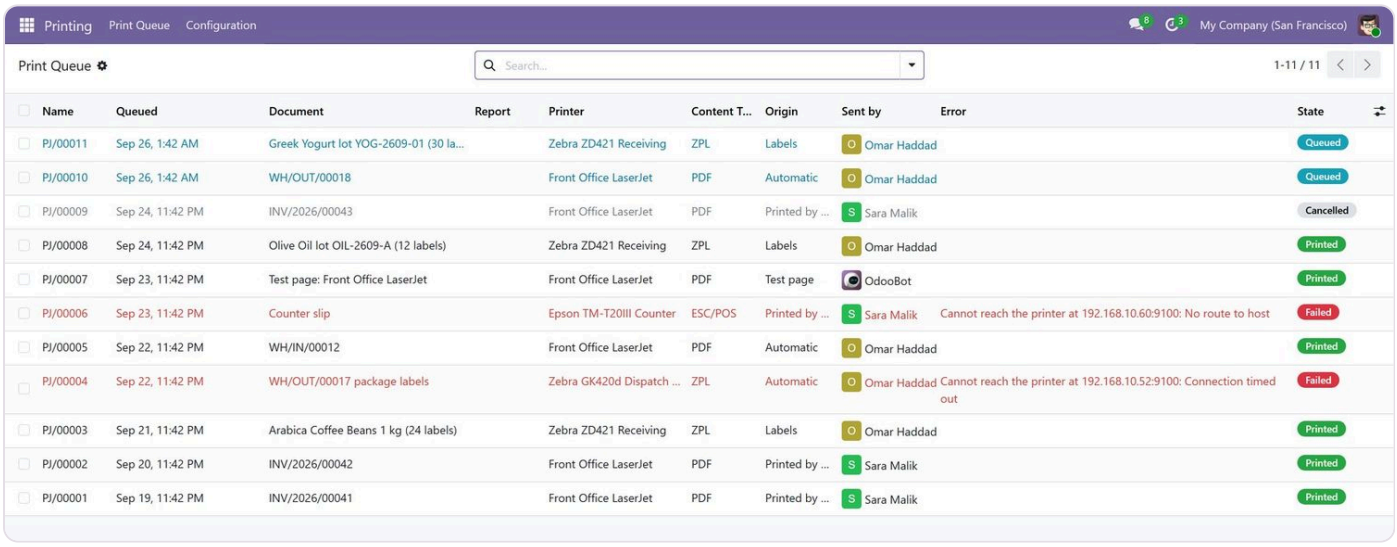
DO IT ON YOUR DATABASE

1. Open Printing, Print Queue
2. Filter on Failed and open one: the reason is in red
3. Press Send again on it once the printer is back

1 The queue

Printing, Print Queue lists each job: number, when it was queued, the document, the printer, the content type (PDF, ZPL, ESC/POS, Raw), where it came from (Printed by a user, Automatic, Labels, Test page), who sent it, the error and the state. Failed jobs are red, waiting ones blue, cancelled ones grey. Select jobs to **Send again** or **Cancel** them together.

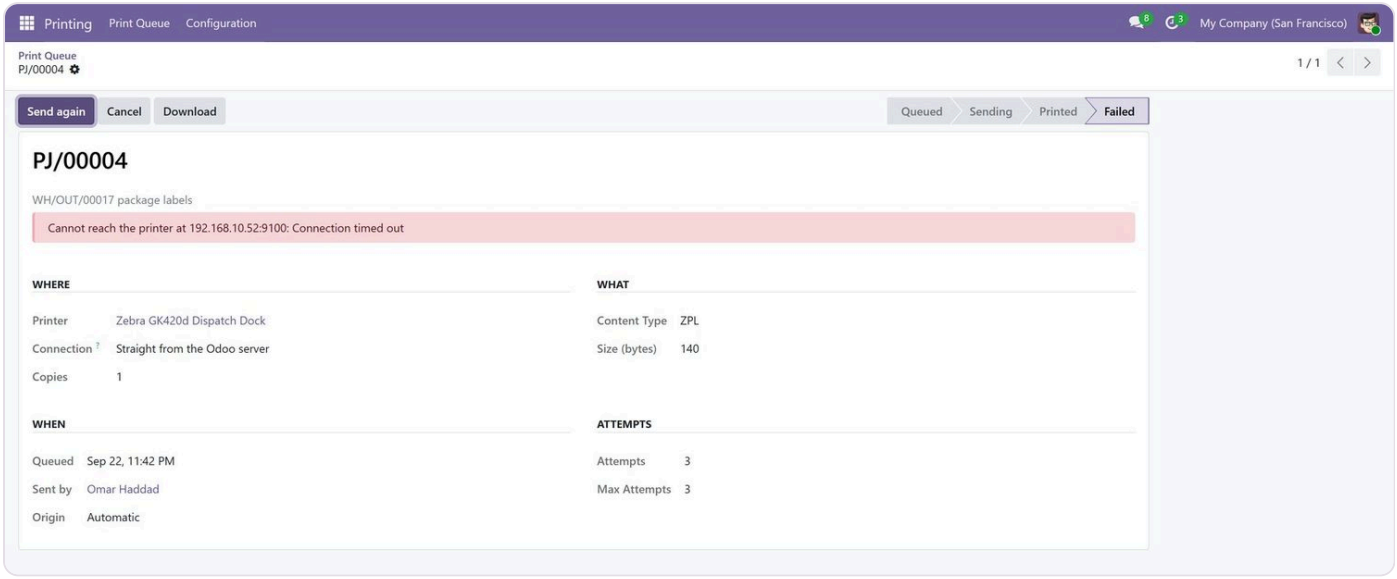
> People with the user right see their own jobs; print managers see every job of the company.



Name	Queued	Document	Report	Printer	Content T...	Origin	Sent by	Error	State
PJ/00011	Sep 26, 1:42 AM	Greek Yogurt lot YOG-2609-01 (30 la...		Zebra ZD421 Receiving	ZPL	Labels	Omar Haddad		Queued
PJ/00010	Sep 26, 1:42 AM	WH/OUT/00018		Front Office LaserJet	PDF	Automatic	Omar Haddad		Queued
PJ/00009	Sep 24, 11:42 PM	INV/2026/00043		Front Office LaserJet	PDF	Printed by ...	Sara Malik		Cancelled
PJ/00008	Sep 24, 11:42 PM	Olive Oil lot OIL-2609-A (12 labels)		Zebra ZD421 Receiving	ZPL	Labels	Omar Haddad		Printed
PJ/00007	Sep 23, 11:42 PM	Test page: Front Office LaserJet		Front Office LaserJet	PDF	Test page	OdooBot		Printed
PJ/00006	Sep 23, 11:42 PM	Counter slip		Epson TM-T20III Counter	ESC/POS	Printed by ...	Sara Malik	Cannot reach the printer at 192.168.10.60:9100: No route to host	Failed
PJ/00005	Sep 22, 11:42 PM	WH/IN/00012		Front Office LaserJet	PDF	Automatic	Omar Haddad		Printed
PJ/00004	Sep 22, 11:42 PM	WH/OUT/00017 package labels		Zebra GK420d Dispatch ...	ZPL	Automatic	Omar Haddad	Cannot reach the printer at 192.168.10.52:9100: Connection timed out	Failed
PJ/00003	Sep 21, 11:42 PM	Arabica Coffee Beans 1 kg (24 labels)		Zebra ZD421 Receiving	ZPL	Labels	Omar Haddad		Printed
PJ/00002	Sep 20, 11:42 PM	INV/2026/00042		Front Office LaserJet	PDF	Printed by ...	Sara Malik		Printed
PJ/00001	Sep 19, 11:42 PM	INV/2026/00041		Front Office LaserJet	PDF	Printed by ...	Sara Malik		Printed

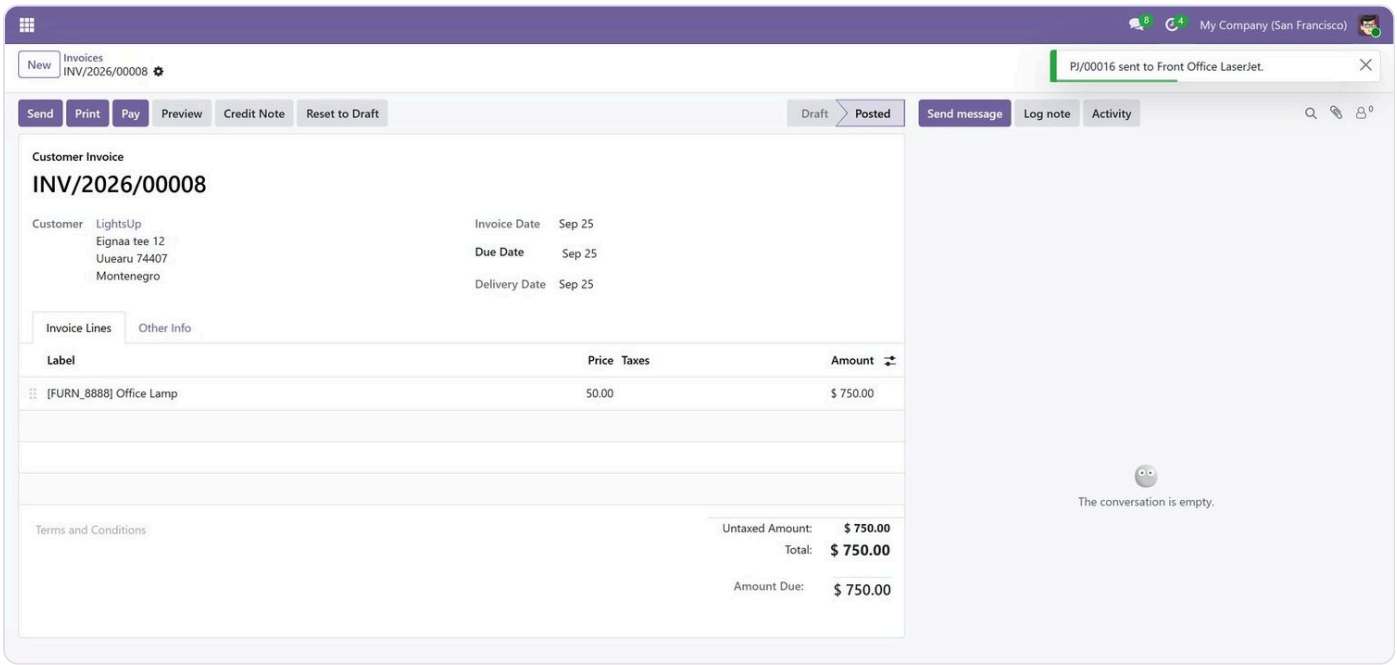
2 A failed job

Open a job to see **Where** (printer, connection, agent, the rule that sent it, copies), **What** (report, content type, size), **When** (queued, sent by, origin, claimed, printed) and **Attempts** (made, allowed, next attempt). **Send again** appears on failed and cancelled jobs, **Cancel** on queued and failed ones, **Download** saves the exact bytes that were sent.



3 What the person printing sees

On the invoice, pressing Print with a matching rule shows a green notice, here PJ/00016 sent to Front Office LaserJet, with the number of copies when there is more than one. No PDF opens.



States and retries

STATE	MEANING
Queued	Waiting for its agent or for the server sender
Sending	Claimed by exactly one sender, which is printing it
Printed	The printer took it
Failed	Every attempt failed; the reason is on the job and on the printer

Cancelled	Stopped by a manager

A failed attempt goes back to the queue after 30 seconds, then 2 minutes, then 10 minutes, and fails for good after three attempts. A job an agent took and never answered for within 10 minutes goes back to the queue as a failed attempt. Two scheduled actions do the work: **Direct Print: send jobs to network printers** every minute (it is also woken as soon as a job is queued) and **Direct Print: requeue stalled jobs, delete old ones** every five minutes, which also deletes printed and cancelled jobs older than the Keep printed jobs setting. A single job is limited to 25 MB.

FILTER OR GROUPING	SHOWS
Waiting / Failed / Printed	Jobs in those states
My jobs / Today	Your own jobs, or today's
Group by Printer / Status / Origin	Counts per printer, per state, per origin

PART 16 Safety and what happens behind the scenes

Printing touches your network and your documents, so the modules are careful about both.

CONCERN	HOW IT IS HANDLED
Who may print what	Reports are rendered as the person printing, with their own access rights. Automatic rules render as the person who validated or confirmed
Agent secrets	The pairing code works once, for 30 minutes. Odoo stores only fingerprints of the code and the token; revoking an agent stops it at once
The firewall	Agents fetch work over HTTPS. Nothing has to be opened inbound
The Odoo server reaching your network	Only the ports in Allowed printer ports. A printer setting cannot be used to reach anything else
Printing twice	A job is claimed by exactly one sender, even with several agents or workers running
A failed print undoing work	Never: the transfer or invoice is saved first, printing follows, and a failure only leaves a chatter note
Old documents left in the database	Printed and cancelled jobs are deleted after the retention you set
Label links	The wizard PDF and ZPL links only open for the person who opened the wizard
Several companies	Templates, printers, agents, rules, workstations and jobs each belong to a company

PART 17 A fifteen minute walk-through

A script for showing the modules to someone, on a test database with the demo data loaded.

MINUTE	SHOW	SAY
0 to 2	Inventory, Configuration, Label Templates	*Six templates come ready: product, variant, lot, package, location and an A4 price tag sheet*
2 to 5	Open the price tag, then Open Designer	*Real data on the canvas. Add a barcode, drag it, nudge it with the arrow keys, Save*
5 to 7	A receipt, Labels	*One label per unit or per line. 12 units, 12 labels; 2.5 kg, one label*
7 to 8	Print PDF, then Show ZPL	*The same design, as a PDF sheet and as Zebra code*
8 to 10	Printing, Configuration, Printers and Print Agents	*One agent in the building serves every printer, and nothing opens on the firewall*
10 to 12	Printing Rules	*Workstation 4, user 2, report 1: the most specific rule wins*
12 to 14	Print an invoice	*No preview: a notice names the job and the printer*
14 to 15	Print Queue, a failed job	*The reason in plain words, and Send again*

PART 18 Questions and fixes

QUESTION OR SYMPTOM	ANSWER
Labels come out too big or too small on the Zebra	The template's ZPL resolution does not match the printer. Set 203, 300 or 600 dpi to what the printer is
PDF labels drift across an A4 sheet	The printer driver is scaling. Print at 100 percent, and check margins and gaps against the label stock
A barcode shows a message instead of bars	The value does not fit the symbology: digits only for EAN and UPC, a correct check digit, plain ASCII for Code 128. The message names the fix
The lot label has no expiry date	The expiry date prints only when Odoo's Expiration Dates setting is on and the lot has one
Print still opens a PDF	No rule matches that report, person and workstation; or Print without preview is off in Preferences; or the person lacks the Direct Printing user right
A printer shows Agent offline	The agent on that computer is not running or cannot reach Odoo. Run `python -m wt_print_agent run --verbose` on it and read the output
A job failed with Cannot reach the printer	The IP address or port is wrong, or the printer is off. Fix it and press Send again
I lost the computer the agent was on	Open the agent and press Revoke. Its token stops working at once
Can it print to a printer at another site	Yes, with an agent at that site. Each site runs its own agent
Is there a monthly fee	No. The agent is part of the module, and there is no cloud print service in between
Does it need Odoo Enterprise	No. All five modules run on Odoo 19 Community and Enterprise

PART 19

The menu map

Inventory, Configuration, Labels

Label Templates	Every label design: size, layout, resolution and its elements
Open Designer	The drag and drop canvas, a button on each template

Print menu on records

Labels (designer)	On products, variants, transfers, lots, packages and locations
Labels button	In the header of every receipt, delivery and internal transfer

Printing (app)

Print Queue	Every job sent to a printer: queued, printed, failed, cancelled

Printing, Configuration

Printers	Each printer, what it prints and how Odoo reaches it
Printing Rules	Which report or label goes to which printer, and when
Label Templates	The same template list, when the Print Suite is installed
Workstations	Desks and stations people print from
Print Agents	The computers in the building that print for Odoo
Settings	Agent download, allowed server ports, how long jobs are kept

Your avatar, Preferences

Printing tab	My workstation, and Print without preview on or off

Want labels and silent printing set up for you?

We design your product, shelf, lot and carton labels to your label stock, set up the print agent on the computers next to your printers, pair your Zebras and office printers, and write the rules so receipts, deliveries and invoices print themselves where they are needed. Then we train the warehouse and the office.

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